General Physics

Forces and acceleration

Understand

Net force is the vector sum of forces acting on an object. In an inertial frame, net force equals mass times acceleration. Acceleration is the rate at which velocity changes.

Worked example

A 2 kg cart has a 10 N force to the right and a 4 N force to the left. Choose right as positive. Net force = 10 - 4 = 6 N. Acceleration = 6 / 2 = 3 m/s^2 to the right.

General Physics

Try it, then check.

Practice question

A 3 kg object experiences a net force of 12 N to the left. Find its acceleration.

Answer and reasoning

4 m/s^2 to the left. Divide net force by mass and keep the direction.

Reflect

Explain the steps in your own words. Create a review card and return to page 1 when you need the original explanation.